CLINICAL · LARYNX & AIRWAY

Benign vocal fold lesions and hoarseness

Most hoarse voices come from the vocal folds themselves, and most of what grows on them is benign: nodules in the teacher, a polyp after a shout, Reinke's oedema in the smoker, a granuloma on the vocal process, papillomas in a child. Benign does not mean harmless. These lesions take voices from those who earn their living by them and airways from small children, and a white patch on a smoker's fold may already be carcinoma in situ. Look carefully, ideally with a stroboscope; give voice therapy its due; and when surgery is needed, preserve the delicate cover of the fold.

AT A GLANCE

- Hoarseness beyond a few weeks needs a look.
- Voice therapy first for nodules.
- Stop smoking and treat reflux.
- Keep the superficial lamina propria safe.

1 CHAPTER ONE Overview

Lesions and tumours.

BENIGN LESIONS AND TUMOURS OF THE LARYNX

NON-NEOPLASTIC (COMMONER)	NEOPLASTIC
Vocal nodules	Squamous papilloma (juvenile and adult)
Vocal polyp	Chondroma
Reinke's oedema	Haemangioma
Contact ulcer and granuloma; intubation granuloma	Granular cell tumour
Leukoplakia and keratosis	Pleomorphic adenoma, oncocytoma
Amyloid	Neurilemmoma, neurofibroma
Cysts: ductal, epidermoid, saccular	Rhabdomyoma, lipoma, fibroma
Laryngocele; sulcus vocalis	

1.1 Assessment of the hoarse voice

Any hoarseness lasting more than three weeks, or sooner in a smoker or with pain, dysphagia, otalgia, stridor or a neck mass, needs laryngoscopy. *Videostroboscopy* shows the mucosal wave, which distinguishes superficial lesions (nodules, polyps) from those that stiffen the fold (cysts, scar, sulcus, carcinoma). Perceptual rating, acoustic analysis and voice handicap indices measure the voice before and after treatment.

1.2 Surgery

Phonomicrosurgery under suspension microlaryngoscopy with cold instruments or laser: a *microflap* is raised through a small incision on the upper surface, the lesion dissected from beneath, and the epithelium laid back, preserving the superficial lamina propria and the vocal ligament. The anterior commissure is never stripped on both sides at once, to avoid a web. Voice therapy before and after prevents recurrence.



FIGURE 1 The set-up for microlaryngeal surgery.
Dhingra, fig. 61.3

OPERATIVE GUIDES IN THEATRE 2 operations

Phonomicrosurgery for benign lesions and office-based laryngeal procedures.

Phonomicrosurgery for Benign Vocal Fold Lesions
Phonomicrosurgery
Office-Based Laryngeal Procedures
Office procedures

CHAPTER TWO

2 Phonotraumatic lesions

Nodules, polyps and Reinke's oedema.

2.1 Vocal nodules

Singers', screamers' or teachers' nodes. *Bilateral, symmetrical* swellings on the free edge at the junction of the anterior third and posterior two-thirds, the midpoint of the membranous fold, where vibration and collision are greatest. They arise from chronic misuse of the voice (shouting, singing too loudly or in the wrong register, speaking at an unnaturally low pitch): submucosal oedema and haemorrhage organise and hyalinise, and the epithelium thickens. Early nodules are soft, red and oedematous; mature ones grey-white and firm. Teachers, singers, actors, vendors and talkative children, especially boys, are typical; in adults, women more often.

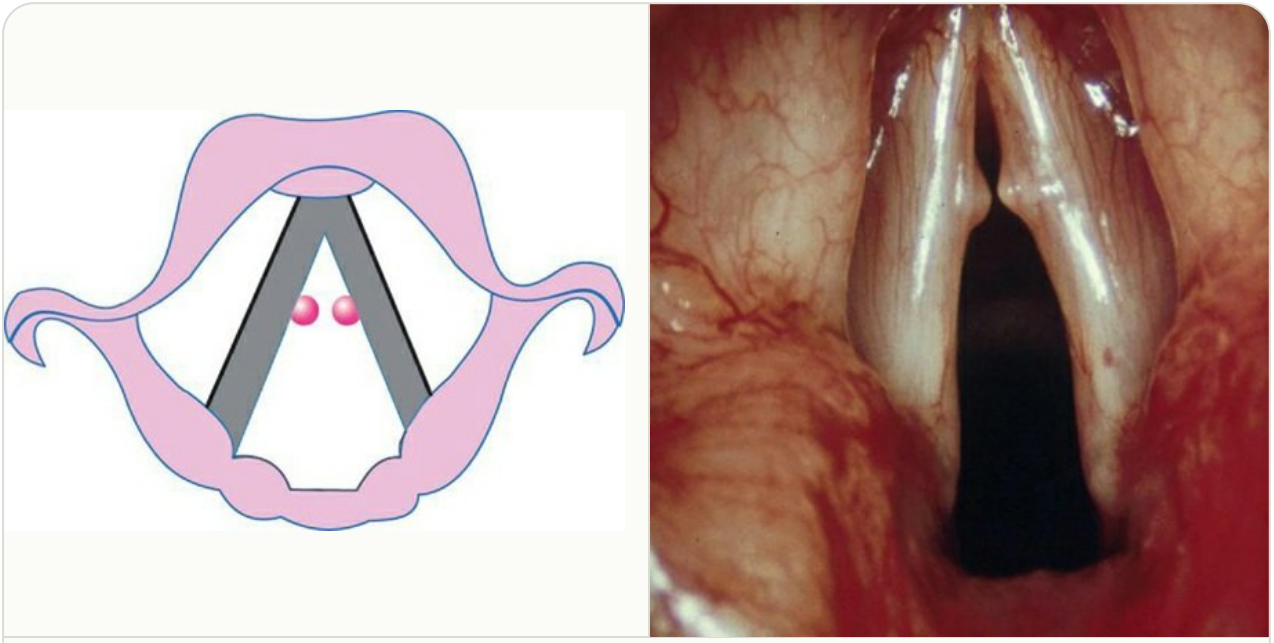


FIGURE 2 First, the site of vocal nodules at the junction of the anterior third and posterior two-thirds of the folds. Second, vocal nodules on laryngoscopy.
Dhingra, figs 61.1 and 61.2

- **Symptoms:** hoarseness, a breathy voice that tires, loss of the upper range in singers, and neck pain after prolonged speaking.
- **Stroboscopy:** an hourglass closure with a preserved mucosal wave.
- **Treatment:** *voice therapy* is the cornerstone; most children's nodules disappear, and many adults'. Lee goes further: lesions that do not shrink after good, compliant therapy are by definition not nodules. Microsurgical excision for large, fibrotic nodules persisting after therapy, with care not to injure the vocal ligament, and continued therapy afterwards; intralesional steroid may help.

2.2 Vocal polyp

Usually a *unilateral*, soft, smooth, often pedunculated swelling at the same site, after sudden shouting causing haemorrhage into the fold, with smoking and allergy contributing; mostly men of thirty to fifty. It may be haemorrhagic, gelatinous or fibrous. Hoarseness, *diplophonia* (two pitches from two differently vibrating folds), and if large and pedunculated, choking or stridor as it flops through the glottis. Treatment is microsurgical excision followed by voice therapy.

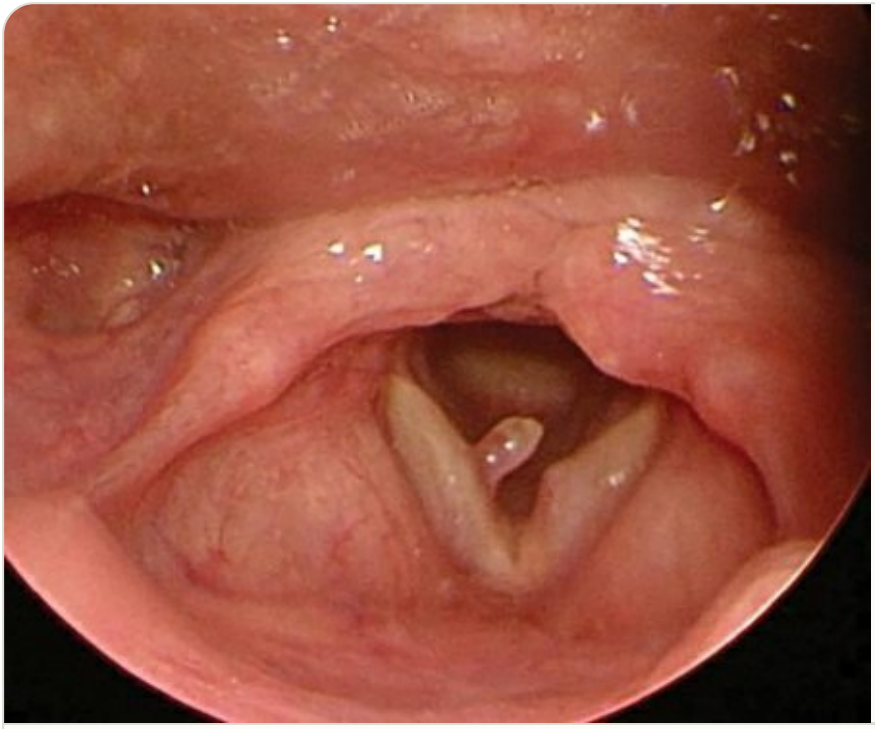


FIGURE 3 A pedunculated polyp on the right vocal fold.
Dhingra, fig. 61.4

2.3 Reinke's oedema (polypoid degeneration)

Bilateral, diffuse, symmetrical, pale, translucent swelling of the whole membranous folds from gelatinous fluid in Reinke's space; smoking is almost always responsible, with voice abuse, reflux and hypothyroidism. The voice becomes deep and rough, more noticeable in women, who may be mistaken for men on the telephone; large swellings obstruct breathing. Stop smoking; treat reflux and thyroid; voice therapy; *decortication* by a lateral incision, suction of the gel and trimming of excess epithelium, traditionally one side at a time.

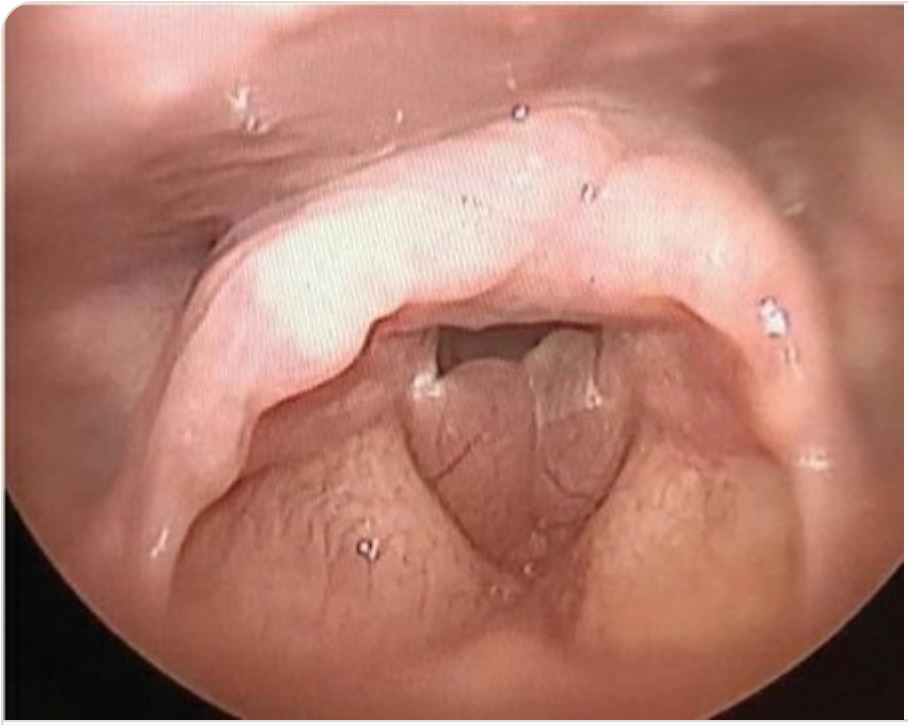


FIGURE 4 Bilateral Reinke's oedema in a 50-year-old woman who smoked.
Dhingra, fig. 61.5

3 CHAPTER THREE Granulomas and keratosis

The back of the fold and the white patch.

3.1 Contact ulcer and granuloma

On the *vocal processes* of the arytenoids, where they hammer together with hard glottal attacks, forceful low-pitched speech and habitual throat clearing;

laryngopharyngeal reflux is a major factor, and intubation another. Hoarseness, pain on speaking radiating to the ear, a foreign-body feeling and constant throat clearing. Ulcers or granulomas sit on one or both vocal processes with red arytenoid mucosa.

- Anti-reflux treatment (twice-daily proton pump inhibitor), even without heartburn.
- Voice therapy to stop throat clearing and correct pitch.
- Inhaled or injected steroid.
- Botulinum toxin to the adductors (lateral cricoarytenoid and thyroarytenoid) for recurrent granulomas.
- Surgery only for airway obstruction, failure of treatment or suspicion of tumour; recurrence after excision is common. Send tissue for histology to exclude carcinoma and granular cell tumour.

3.2 Intubation granuloma

Injury to the vocal processes from rough intubation, a large tube or prolonged intubation; usually bilateral on the posterior folds, with hoarseness and, if large, dyspnoea. Many mature into a pedunculated mass and drop off; treatment is as for contact granuloma.

3.3 Leukoplakia and keratosis

White plaques or warty thickening on the upper surface of one or both folds with normal mobility, in smokers. Histology ranges from simple keratosis through dysplasia to carcinoma in situ and invasive carcinoma, so it is a *precancerous condition*. Biopsy or *excisional microflap stripping* for histology, stop smoking, treat reflux, and follow up for recurrence and new lesions; avoid repeated deep stripping that scars the voice.

THE WHITE PATCH ON A SMOKER'S FOLD

A keratotic or red lesion with a reduced mucosal wave on stroboscopy, irregular surface, or impaired mobility is suspicious for invasive carcinoma: biopsy rather than repeated observation.

3.4 Amyloid

Men of fifty to seventy: a yellow or orange submucosal mass in the false fold, ventricle, true fold or subglottis, causing hoarseness or dyspnoea; apple-green birefringence with Congo red. Exclude myeloma and systemic amyloidosis; remove localised deposits endoscopically; the outlook is good.

4 CHAPTER FOUR Cysts and sulcus

Hidden in the fold.

4.1 Ductal (retention) cysts

From blocked seromucinous glands in the vallecula, aryepiglottic fold, false fold, ventricle and piriform fossa: silent when small; hoarseness, cough, throat pain and dyspnoea when large.

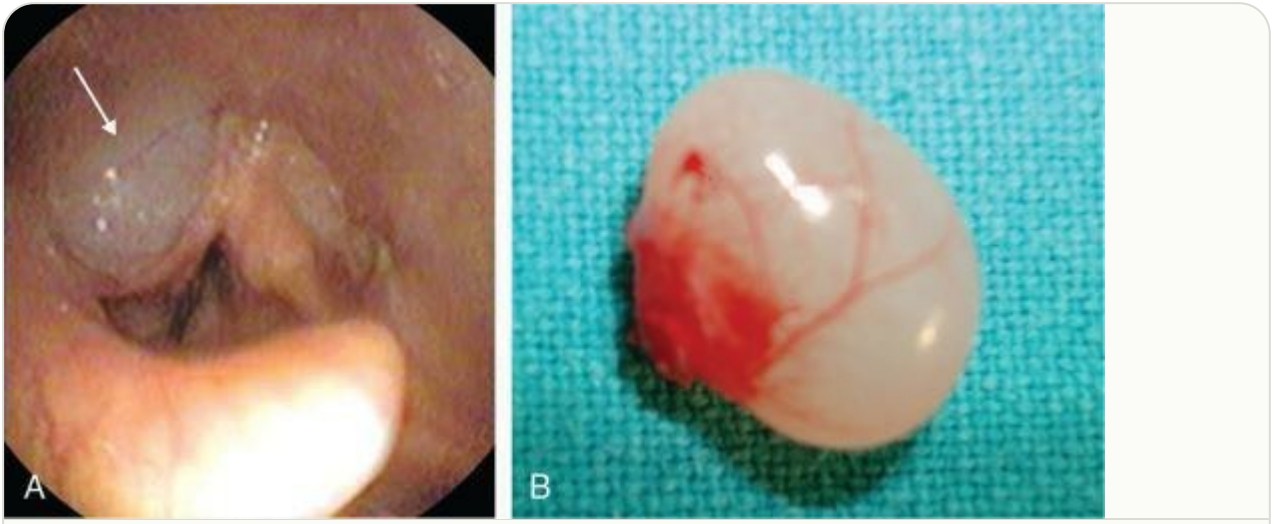


FIGURE 5 (A) An aryepiglottic cyst causing intermittent laryngeal obstruction. (B) The cyst after removal.
Dhingra, fig. 61.6

4.2 Vocal fold cysts

- **Epidermoid cyst:** an inclusion cyst of keratin, congenital or from phonotrauma, white and deep in the fold, often against the ligament.

- **Mucous retention cyst:** smooth, glistening and fluid-filled.

Usually unilateral, sometimes with a reactive nodule on the opposite fold, and easily confused with nodules; stroboscopy shows a stiff segment with reduced mucosal wave, and they do not resolve with voice therapy. Microflap excision, ideally intact; residual wall recurs.



FIGURE 6 An epidermoid cyst of the left vocal fold, full of cheesy material at excision.
Dhingra, fig. 61.7

4.3 Saccular cysts and laryngocele

Obstruction of the saccule's mouth fills it with mucus (a *saccular cyst*: anterior, bulging from the ventricle; or lateral, into the false fold and aryepiglottic fold and even the neck). Distension with air forms a *laryngocele*: internal, external (through the thyrohyoid membrane, a reducible neck swelling that grows with coughing or Valsalva) or combined, linked to raised pressure in wind players, glass blowers and weight lifters. Diagnosed by laryngoscopy, radiographs with Valsalva and CT. Endoscopic marsupialisation or excision for internal lesions; external excision for external and combined lesions. In an adult, a ventricular carcinoma obstructing the saccule must be looked for and biopsied.

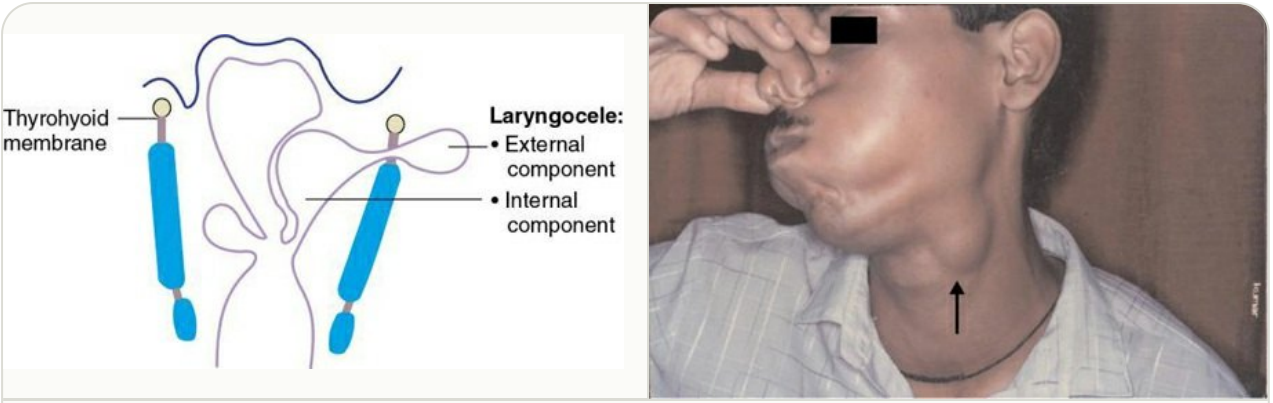
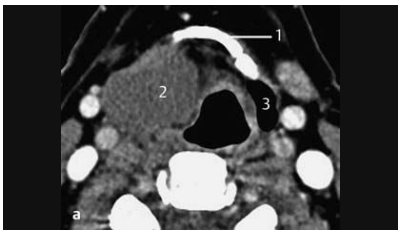


FIGURE 7 First, a laryngocele of the mixed type with internal and external components. Second, a left laryngocele as seen on Valsalva (arrow).

Dhingra, figs 61.8 and 61.9

IMAGING IN THE APP 1 case

Laryngocele on imaging.



Laryngocele

Infection & inflammation · 10 images

4.4 Sulcus vocalis

A furrow along the free edge where the lamina propria is deficient and the epithelium dips to the ligament or muscle: congenital, or from phonotrauma, perhaps an opened epidermoid cyst. A breathy, weak voice with incomplete closure, a spindle-shaped gap and a reduced mucosal wave; confirmed by palpation at microlaryngoscopy. Prolonged voice therapy first; treat reflux and allergy. Surgery has limited success: steroid or filler injection, fat or fascia implantation, excision of the sulcus, slicing or multiple relaxing incisions, undermining, and medialisation thyroplasty for closure.

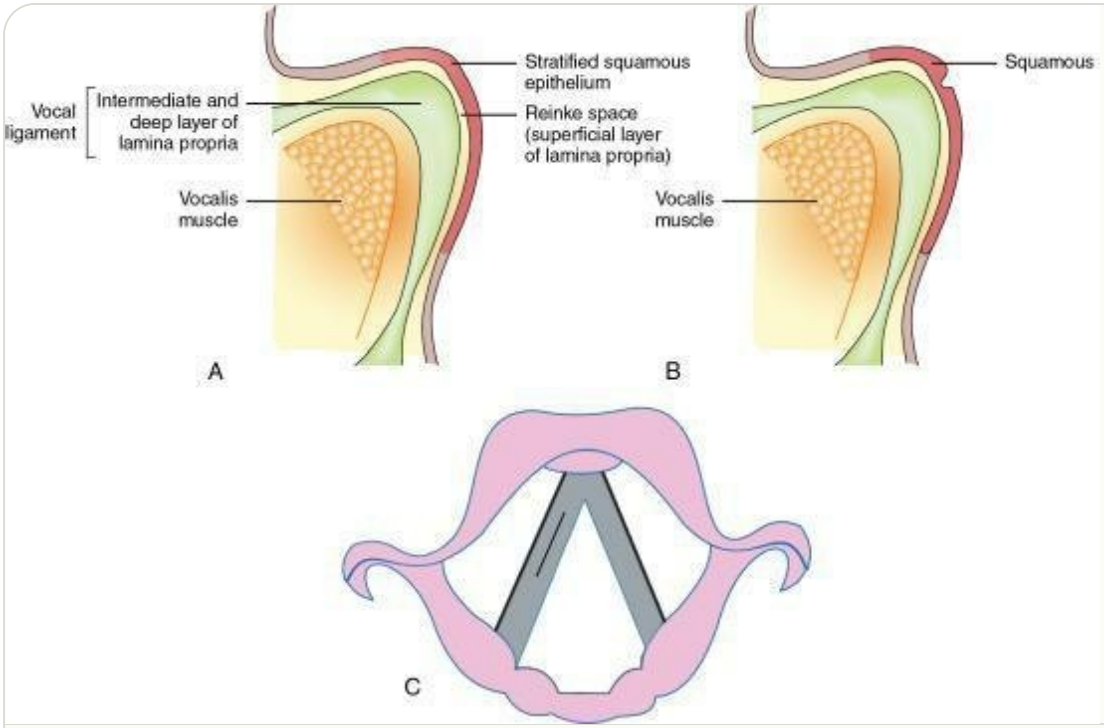


FIGURE 8 (A) The layers of a normal vocal fold. (B) A sulcus vocalis in coronal section. (C) Its appearance on laryngoscopy.

Dhingra, fig. 61.10

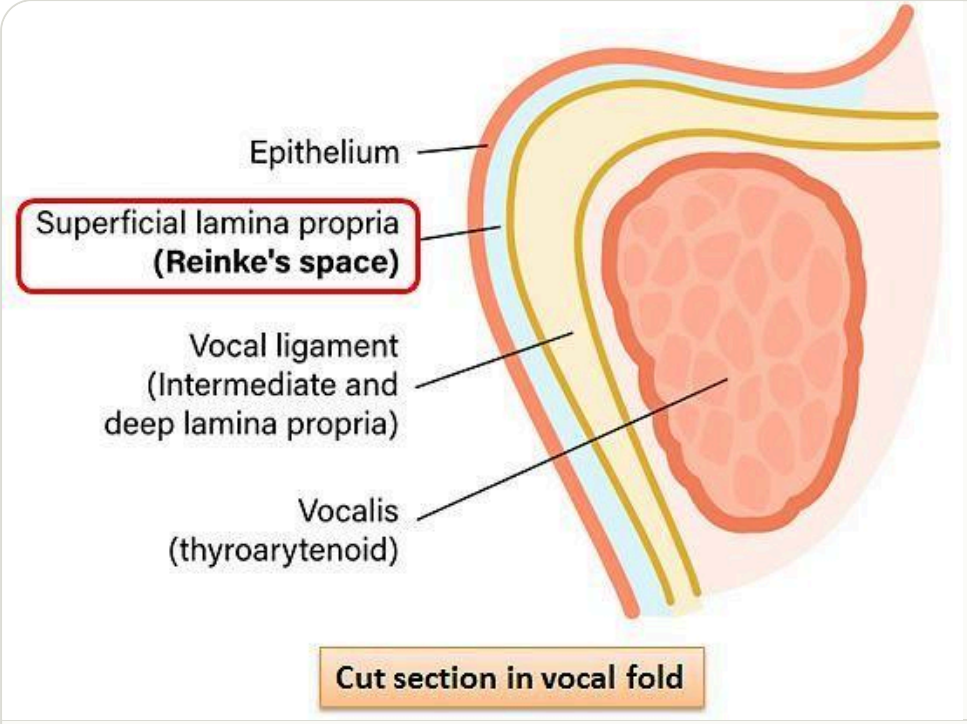


FIGURE 9 The layers of the vocal fold: Reinke's space and the vocal ligament.

Anatomy atlas · ENT Steps p. 246

5 CHAPTER FIVE Neoplasms

Papillomas first.

5.1 Recurrent respiratory papillomatosis

The commonest benign tumour of the larynx (about 80% of benign neoplasms) and of the child's larynx, caused by *HPV 6 and 11*.

- **Juvenile onset:** acquired from mothers with genital warts at birth; first-born children of young mothers by vaginal delivery are at greatest risk (caesarean section does not reliably prevent it). Presents at two to five years with hoarseness, a weak cry or aphonia and then stridor; often misdiagnosed as asthma or croup. Multiple, aggressive and recurrent, at junctions of squamous and ciliated epithelium (supraglottis, glottis, subglottis), spreading to the trachea and bronchi, especially after tracheostomy.
- **Adult onset:** men of thirty to fifty, often single, at the anterior fold or commissure, less aggressive; malignant change is rare (more with HPV 16 and 18, smoking and irradiation).

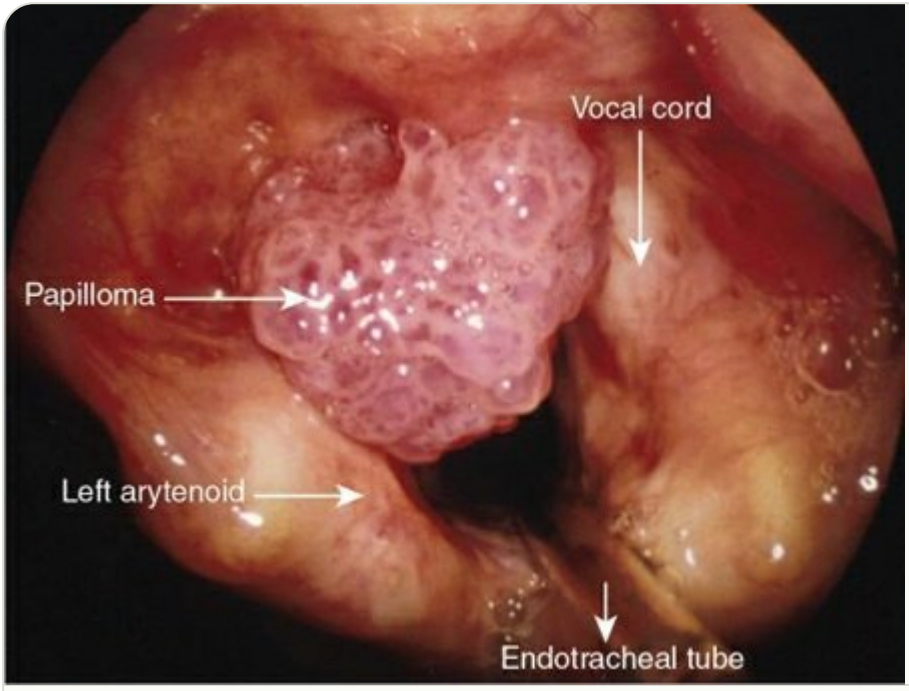


FIGURE 10 Supraglottic papillomatosis.
Dhingra, fig. 61.11

5.1.1 Treatment

The aim is a safe airway and a useful voice with as few scars as possible; surgery debulks, it does not cure.

- **Surgery:** microdebrider, CO2 laser or pulsed KTP laser under suspension laryngoscopy, avoiding injury to the vocal ligament and the anterior commissure; repeated as often as needed. Office KTP laser in adults.
- **Adjuvants** for frequent recurrence or distal spread: intralesional *cidofovir*, systemic or intralesional *bevacizumab*, indole-3-carbinol, and (older) interferon alpha and retinoids; the HPV vaccine as therapy is under study, and as prevention will reduce maternal transmission.
- **Tracheostomy** is avoided if possible, because it encourages tracheal spread.

OPERATIVE GUIDES IN THEATRE 1 operation

Recurrent respiratory papillomatosis.

Recurrent Respiratory Papillomatosis
Papillomatosis

5.2 Chondroma

Usually from the posterior lamina of the cricoid, also the thyroid, arytenoid or epiglottis; men of forty to sixty. A slow, hard, submucosal mass causing dyspnoea when subglottic, or a lump feeling and dysphagia when growing back into the hypopharynx. CT shows a calcified cartilaginous mass; biopsy (with the laser through hard tissue) to exclude chondrosarcoma. Excision through a laryngofissure or lateral pharyngotomy, preserving as much cricoid as possible; laryngectomy for large or recurrent tumours.

5.3 Haemangioma

- **Infantile:** subglottic, capillary, presenting with stridor in the first six months, half with skin haemangiomas; involutes, and is treated with propranolol (see congenital lesions).
- **Adult:** cavernous, on the fold or supraglottis; left alone if silent, treated with steroids, laser or excision if symptomatic.

5.4 Granular cell tumour

From Schwann cells, submucosal, usually on the posterior fold; the overlying *pseudoepitheliomatous hyperplasia* can be mistaken for well-differentiated carcinoma. Complete excision prevents recurrence.

5.5 Others

Pleomorphic adenoma, oncocytoma, neurilemmoma and neurofibroma, rhabdomyoma, lipoma and fibroma are rare and excised endoscopically or through an external approach.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Site of nodules** and why there.
- **Nodule versus polyp versus cyst:** bilaterality, stroboscopy and response to therapy.
- **Reinke's oedema** and smoking.
- **Contact granuloma:** reflux and voice misuse.
- **Leukoplakia** as a premalignant lesion.
- **Laryngocele:** types and the hidden carcinoma.
- **Recurrent respiratory papillomatosis:** HPV types, behaviour, treatment and adjuvants, and why tracheostomy is avoided.

6.2 Common pitfalls

COMMON PITFALLS

- Operating on nodules before a proper course of voice therapy.
- Stripping both sides of the anterior commissure at once.
- Repeatedly observing a white patch in a smoker without a biopsy.
- Diagnosing a child with papillomas as asthma.

Written for this app from Oxford Textbook of Otolaryngology ch. 47, 71, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 12, Dhingra ch. 61 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Laryngocele

IMAGING FINDINGS

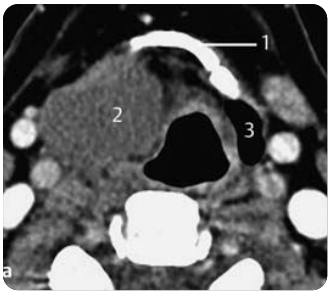
- Air or fluid-filled sac from the laryngeal ventricle
- Internal (paraglottic) or external through the thyrohyoid membrane
- Rim enhancement if infected (pyolaryngocele)

TOP DIFFERENTIALS

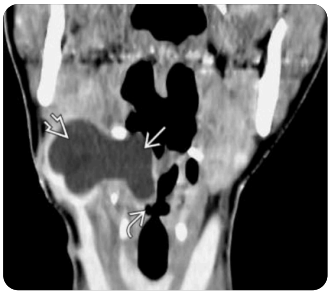
- Thyroglossal duct cyst
- Second branchial cleft cyst
- Saccular cyst

CHECKLIST

- Exclude a carcinoma at the ventricle in adults
- Airway compromise



CT with intravenous contrast enhancement, axial. The axial slice at the lower border of the hyoid (1) reveals a homogeneous, expanding…



Coronal CECT reformat shows a similar lesion. This is also known as a "mixed" laryngocele because it contains internal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 292.